Beyond the Top Pose: Posterior Calibration and Trustworthiness in Generative Molecular Docking

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Motivation & Scientific Justification

Molecular docking predicts how a small drug-like molecule (a ligand) binds to a target protein, which then forms a so called complex. It is one of the earliest computational filters in structure-based drug discovery, a pipeline that takes roughly a decade, costs on the order of a billion dollars, and can fail at any stage [1]. With an estimated 10^{60} drug-like molecules to consider [2], docking is what lets researchers screen enormous virtual libraries before committing to expensive experiments.

The field has recently pivoted from classical physics-based search [3] toward deep generative (diffusion) models such as DiffDock [4] and SigmaDock [5]. Classical methods slowly search over poses and rank them with a hand-crafted scoring function, which approximates binding energy (van der Waals, electrostatics etc.) that is only a rough proxy for reality. Rather than optimising such a scoring function, diffusion models learn to sample from a probability distribution over plausible binding poses. They are judged almost entirely on *accuracy*: how often the single best-ranked pose matches the experimentally determined structure. Accuracy alone, however, hides a dangerous blind spot. A model can be confidently wrong, concentrating all its predicted poses around an incorrect answer while reporting no doubt. In a high-stakes, failure-prone pipeline, a confidently wrong prediction is far more damaging than an honestly uncertain one. The decisive question is therefore not only "how often is the model right?" but "how trustworthy is its output?"

This project asks whether the pose distributions produced by generative docking models are trustworthy beyond headline accuracy, and what determines that trustworthiness: the model's underlying architecture, or the numerical sampler used to generate poses at inference time.

Methodology

Trustworthiness was decomposed into two measurable properties beyond raw accuracy. The first, *physical validity*, asks whether a generated pose is chemically and sterically plausible, which is assessed with the established PoseBusters suite [6]. The second, *posterior calibration*, asks whether the spread of predicted poses honestly reflects the model's uncertainty, which is whether the distribution assigns the right probability mass across the space of possible answers. It measures the extent to which the true data-generating process matches the learned Posterior.

Calibration is the harder and more novel axis. Because generative docking models provide no tractable likelihood, and there is only a single known ground truth rather than an analytical distribution, standard statistical tests do not apply. Two recent likelihood-free diagnostics, the Test of Accuracy with Random Points (TARP) [7] and the MIRA score [8], were adapted to the geometry of docking (Fig. 1), where a pose lives on a manifold of translations, rotations, and torsion angles rather than in ordinary Euclidean coordinates. This adaptation is the project's first technical contribution [1].

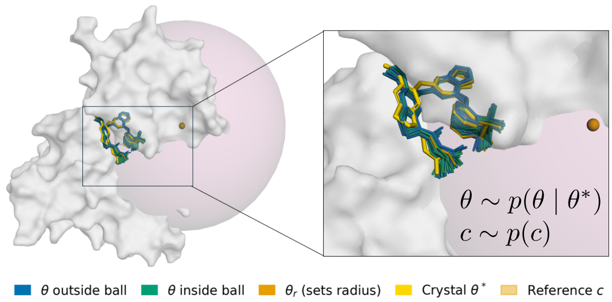

Figure 1: The MIRA calibration test adapted to docking. A spherical test region (pink) is anchored at a random centre, its radius set by a reference pose drawn from the model itself. The test counts how often the true crystal pose (yellow) falls inside, and how many samples in the generated ensemble fall inside the ball. Aggregated over many centres and complexes, this measures whether the model's posterior spreads probability mass honestly rather than overconfidently.

The second contribution opens up the sampling process itself. A trained diffusion model can be integrated in many ways, and the choice of solver is made entirely at inference time, independently of training [9]. The study compared integration of the deterministic ordinary differential equation (ODE) against the stochastic differential equation (SDE), and varied the number of refinement steps, for two state-of-the-art models across multiple datasets and protein families. Together, these let reliability be attributed to its architectural and inference-time sources.

Key Findings

Validity is architectural, not fixable by ranking. For DiffDock, roughly a third of complexes yield no valid pose at all across forty samples, a hard ceiling that no smarter re-ranking can lift. Substituting alternative rankers barely moves validity, whereas SigmaDock's fragment-based design produces valid geometry directly. The ranker cannot find a good pose in the ensemble if the model never generated one.

Both models are systematically overconfident. Across every dataset, protein family, and sampling regime tested, calibration scores fell below the value expected of an honest posterior (Fig. 2): the models concentrate their predictions too tightly, even when they locate the correct binding pocket. Critically, this failure is independent of accuracy and validity, exposing three genuinely separate axes of model quality, of which a conventional leaderboard reports only the first.

Calibration is set first by architecture, then by the sampler. SigmaDock's well-conditioned, decoupled representation of the pose space is markedly better calibrated than DiffDock's entangled one, and its advantage concentrates precisely on the torsional degrees of freedom, its design was built to disentangle, indicating strong evidence that the gain is structural rather than incidental. The sampler is a real but secondary lever: a ten-step stochastic sampler gives SigmaDock

[1] The Codebase and the full report can be found on GitHub and the data on this Drive.

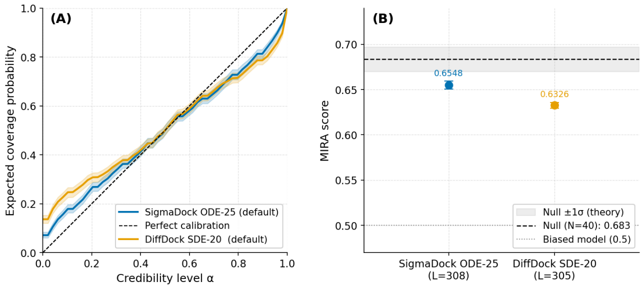

Figure 2: Calibration result. (A) Expected-coverage (TARP) curves: a perfectly calibrated model follows the diagonal; both models show an S-curve, signalling overconfidence, but SigmaDock (blue) tracks it more closely than DiffDock (orange). (B) The corresponding MIRA scores: both fall below the honestly-calibrated baseline (dashed), with SigmaDock notably higher.

its best calibration and its fastest runtime, at a cost of only about two accuracy points. This is a configuration that an accuracy-only search would never surface, and one where the stiffer DiffDock instead collapses.

Recommendation & Next Steps

The central message is that trustworthiness in generative docking is invisible to accuracy-only evaluation, and that the field's habit of tuning and ranking models on a single top-1 number actively rewards the mode-seeking samplers that suppress honest uncertainty. This work argues for *calibration-aware docking*: training objectives that target both distributional coverage and point accuracy, sampler-aware ranking, and adoption of the MIRA score as a practical criterion for Bayesian model selection. For any application that consumes the full posterior, rather than the single top pose, we showed that SigmaDock with a ten-step stochastic sampler is the best default.

A concrete next step follows from the observation that both models still use only first-order numerical integration. Higher-order solvers now mature in image generation, Heun predictor–correctors [10] and DPM-Solver schemes [11, 12], may let even stiff architectures reach calibrated coverage at the low step counts that currently break them. These directions extend naturally to the newer joint and co-folding docking regimes now redefining the field [13].

Research Impact & Conclusion

The impact of this work is twofold. Methodologically, the adapted TARP and MIRA diagnostics give the docking community a transferable, likelihood-free way to measure posterior reliability, a tool it currently lacks. Practically, the finding that architecture, not re-ranking, sets both validity and calibration tells developers where to invest, and gives practitioners a principled basis for choosing models and samplers when the full distribution of poses is what matters for the decision at hand. As generative models increasingly drive early-stage drug discovery, knowing when to trust them is as important as knowing how often they are right [2].

[2] LATEX word count 997/1000

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